Notes: Haddad, Huebner, and Loualiche (AER, 2025)

How Competitive Is the Stock Market? Theory, Evidence from Portfolios, and Implications for the Rise of Passive Investing

Contents

1 Big picture

- **Question.** How competitive is the stock market in the sense that investors strategically adjust their trading aggressiveness when other investors become less aggressive, for example when assets move from active to passive management?
- **Core object.** The paper estimates the **degree of strategic response**, denoted χ: how much an investor's demand elasticity responds to the aggregate elasticity provided by other investors in a stock.
- **Conventional benchmark.** The usual efficient-markets intuition is that if some investors stop searching for mispricing, other investors step in and fully replace them, so asset prices and aggregate demand elasticities do not change.
- **Main finding.** Strategic responses are large but incomplete. The baseline estimate is $\chi = 2.97$, implying that active investors offset about two-thirds of the direct impact of a rise in passive investing.
- **Key quantitative implication.** Because strategic responses are incomplete, the rise of passive investing over 2001–2020 made stock-level demand curves about 11 percent more inelastic.
- **Data and method.** The authors estimate an institutional demand system using detailed 13F portfolio holdings, prices, stock characteristics, and instruments for both price and aggregate elasticity.
- **New empirical challenge.** The aggregate elasticity is endogenous and jointly determined with investor elasticities. The paper solves this by using cross-stock variation in the mix of competing investors and by constructing instruments from lagged AUM and investment universes.
- **Main equilibrium idea.** There are two layers of equilibrium. First, prices clear the market given investors' demand curves. Second, investors choose demand elasticities in response to the aggregate elasticity created by all investors.
- **Broader implication.** Market structure matters. The composition of investors affects price elasticity, volatility, informativeness, and liquidity because strategic responses do not fully neutralize changes in investor behavior.

2 Incremental contribution of the paper

2.1 Contribution relative to efficient-market intuition

- A common view in finance is that competition among investors is so strong that changes in one group of investors do not affect prices: if passive investors stop trading on information, active investors simply trade more aggressively.
- This paper turns that informal intuition into a measurable equilibrium parameter, χ.
- It shows that the replacement force is real but partial. Investors react to slack left by others, but not enough to fully restore the original aggregate elasticity.

Interpretation: The incremental contribution is not to claim that investors do not compete. It is to quantify how much competition actually offsets changes in investor behavior.

2.2 Contribution relative to empirical demand systems

- Koijen and Yogo-style demand systems estimate investor demand curves and price elasticities, usually treating each institution's elasticity as an individual characteristic.
- This paper adds a second equilibrium layer: investors' elasticities are themselves strategic choices affected by the aggregate elasticity of the stock.

- Because individual and aggregate elasticities are jointly determined, one cannot estimate each investor separately. The paper estimates the whole system simultaneously.

Interpretation: The paper moves empirical demand estimation from partial equilibrium demand curves to a strategic equilibrium among institutions.

2.3 Contribution relative to passive investing studies

- Prior work documents that passive ownership affects prices, volatility, liquidity, or price informativeness.
- This paper supplies a mechanism: passive investing lowers aggregate demand elasticity directly, while active investors partly offset the effect through strategic response.
- The estimated pass-through translates the growth of passive investing into a measurable change in the slope of individual stock demand curves.

Interpretation: The paper does not simply ask whether passive investing matters. It decomposes why it matters and how much active investors undo the direct effect.

2.4 Contribution relative to limits-to-arbitrage theories

- Theoretical models emphasize frictions that limit arbitrage: costly information, mandates, incentives, bounded rationality, cognitive constraints, and market power.
- The paper remains semistructural and does not take a stand on which friction dominates.
- Instead, it estimates the net equilibrium strength of strategic response, aggregating all these frictions into one central object.

Interpretation: This is useful because the empirical pass-through of passive investing depends on the total response of the active sector, not on a single microfoundation.

3 Model and notation

3.1 Two-layer equilibrium model

The paper summarizes the model in Table 1. The first layer is price clearing. Investor i's demand for a stock is locally log-linear:

$$d_i = \bar{d}_i - \mathcal{E}_i(p - \bar{p}), \tag{1}$$

where d_i is the log position, p is log price, and $\mathcal{E}_i$ is the investor's demand elasticity. The stock price clears the market:

$$\int D_i(p)\, di = S. \tag{2}$$

The aggregate elasticity is the position-weighted average of investor elasticities:

$$\mathcal{E}_{agg} = \frac{\int \mathcal{E}_i D_i\, di}{\int D_i\, di}. \tag{3}$$

Interpretation: An investor with a high $\mathcal{E}_i$ trades more aggressively: she buys much more when the stock becomes cheap.

3.2 Second layer: strategic choice of elasticity

The second layer describes how investors choose elasticities in response to others:

$$\mathcal{E}_i = \bar{\mathcal{E}}_i - \chi \mathcal{E}_{agg}. \tag{4}$$

- $\bar{\mathcal{E}}_i$ is the investor-specific component of elasticity.
- χ is the degree of strategic response.
- If $\chi > 0$, investors become more elastic when the aggregate elasticity provided by others is lower.
- If $\chi = 0$, investors do not strategically respond to other investors' behavior.
- If $\chi \to \infty$, investors respond so strongly that aggregate elasticity is effectively pinned down.

Interpretation: The sign convention means that when other investors become less aggressive and $\mathcal{E}_{agg}$ falls, a given investor raises her own elasticity.

3.3 Pass-through of passive investing

When some investors become passive, their demand elasticity effectively falls to zero. The direct effect lowers aggregate elasticity. Active investors then respond. A useful approximate pass-through formula is:

$$\text{Pass-through} \approx \frac{1}{1 + \chi \times ActiveShare}. \tag{5}$$

Using the average active share of about 68 percent and $\chi \approx 3$, the pass-through is about

$$\frac{1}{1 + 3 \times 0.68} \approx 0.33. \tag{6}$$

Interpretation: Active investors offset about two-thirds of the direct effect of a rise in passive investing, but one-third remains. That remaining effect makes stock demand curves more inelastic.

4 Data and measurement

4.1 Institutional holdings

- The empirical analysis uses institutional holdings from 13F filings over 2001:Q1–2020:Q4.
- The cross section includes many institutions and stocks, with institutional positions measured at quarterly frequency.
- Institutions are grouped by investor type and AUM in some specifications to ensure enough observations for estimating elasticities.
- Stock characteristics include market capitalization, book equity, profitability, investment, dividend yield, and related variables.

Interpretation: Institutional holdings are the core data because the model requires observed portfolio weights for many investors across many stocks.

4.2 Active and passive investors

- The authors classify the fraction of active investors stock-by-stock using the estimated model.
- Passive strategies are treated as having zero price elasticity in the demand system.
- The median active share across stocks falls from 81.4 percent in 2000:Q4 to 59.5 percent in 2020:Q4.

Interpretation: The decline in active share is the empirical force used to quantify how passive investing changes aggregate elasticity.

4.3 Aggregate stock-level elasticity

For each stock k and date t, the model produces an aggregate elasticity $\mathcal{E}_{agg,k,t}$. This elasticity measures how strongly total demand for the stock responds to its price.

- A higher elasticity means that small price changes induce larger demand responses.
- A lower elasticity means stock demand is more inelastic, so trading shocks have larger price effects.
- The paper estimates substantial cross-sectional and time-series variation in aggregate stock-level elasticity.

Interpretation: Aggregate elasticity is the bridge between investor behavior and asset prices.

5 Empirical specification and identification

5.1 Demand system

The empirical demand system follows the spirit of Koijen and Yogo but adds strategic response. A simplified version is:

$$\log\left(\frac{w_{ik}}{w_{i0}}\right) - p_k = \delta_{0i} + \delta_{1i}' X_k - \mathcal{E}_{ik} p_k + \xi \mathcal{E}_{aggk} + \zeta' \mathcal{E}_{aggk} X_k + \varepsilon_{ik}, \tag{7}$$

where w_{ik} is institution i's portfolio weight in stock k, w_{i0} is the outside asset weight, X_k are stock characteristics, and p_k is price.

The strategic elasticity component is:

$$\mathcal{E}_{ik} = \mathcal{E}_{0i} + \mathcal{E}_{1i}' X_k - \chi \mathcal{E}_{aggk}. \tag{8}$$

Interpretation: The coefficient χ is estimated from how a given investor's elasticity varies across stocks with different surrounding investor mixes.

5.2 Reflection problem

A high aggregate elasticity could arise because:

1. each investor is intrinsically elastic; or
2. investors are responding to the elasticities of others.

This is a reflection problem. The paper solves it using cross-stock variation: the same institution faces different sets of competitors in different stocks.

Interpretation: Comparing the same institution across stocks with different competing investors separates investor traits from strategic response.

5.3 Instruments

The paper instruments both price and aggregate elasticity. Instruments include:

- lagged AUM and investment universe measures;
- characteristics of other investors' portfolios;
- a constructed instrument for aggregate elasticity obtained by solving a version of the two-layer model using predetermined investor characteristics.

Interpretation: Instruments are necessary because price and aggregate elasticity are equilibrium objects jointly determined with demand.

5.4 Relevance diagnostics

Figure 2 reports first-stage relevance diagnostics. The Kleibergen-Paap F-statistics are generally above conventional weak-instrument thresholds for both price and the price-by-elasticity interaction.

Interpretation: The instruments have enough power to identify strategic response in the demand system.

6 Main estimates and implications

6.1 Degree of strategic response

The baseline estimate is:

$$\hat{\chi} = 2.97. \tag{9}$$

Across robustness specifications, the estimate usually lies between about 2.30 and 3.27. This means active investors do respond strongly when other investors are less aggressive, but not strongly enough to fully undo the initial change.

Interpretation: The estimate rejects both extremes: no strategic response and perfectly competitive replacement.

6.2 Aggregate elasticity across stocks

The estimated aggregate elasticity is lower for larger stocks. This is counterintuitive if one expects large stocks to be more efficiently priced, but it is consistent with the idea that institutions face constraints in large positions and that large-cap stocks occupy a large share of portfolios.

Interpretation: Demand elasticity depends not only on information efficiency, but also on the portfolio constraints and incentives of institutions.

6.3 Rise of passive investing

The active share of investors falls sharply over 2001–2020. With $\chi \approx 3$, strategic responses absorb about two-thirds of the direct effect. The remaining one-third produces an 11 percent decline in aggregate demand elasticity.

Interpretation: Passive investing matters even in an active market. The competitive response is strong but incomplete.

6.4 Asset-price implications

More inelastic stock demand is associated with:

- higher volatility;
- lower price informativeness in some specifications;
- higher illiquidity in some specifications;
- larger price impact of order-flow shocks.

Interpretation: The composition of investors affects not only asset ownership but also the price formation process.

7 Tables and figures

7.1 Table 1 and Figure 1: two-layer model and identification logic

Read:

- Table 1 summarizes the model: investor demand and elasticity decisions in the first layer, and market clearing plus aggregate elasticity in the second layer.
- Figure 1 illustrates the identification strategy with investors holding multiple stocks and facing different competitor mixes.
- The same investor can be compared across stocks, allowing the authors to separate individual elasticity from response to others.

Economic message: The paper's central innovation is to estimate strategic response using portfolio variation across stocks and investors.

TABLE 1—THE TWO-LAYER MODEL OF ASSET MARKET

	Individual decision	Equilibrium condition
Demand	$d_i = \underline{d}_i - \mathcal{E}_i \times (p - \bar{p})$	$\int D_i(p) = S$
Elasticity	$\mathcal{E}_i = \mathcal{E}_i - \chi \times \mathcal{E}_{agg}$	$\int \mathcal{E}_i D_i / S = \mathcal{E}_{agg}$

Table 1: Two-layer model of the asset market

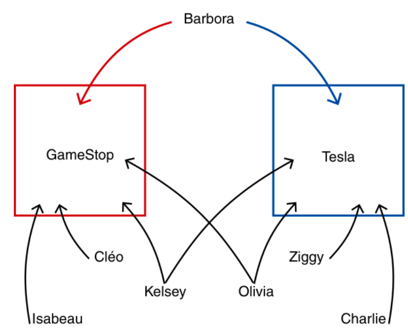

FIGURE 1. ILLUSTRATION OF IDENTIFICATION STRATEGY

Figure 1: Identification strategy

7.2 Figure 2 and Table 2: instruments and strategic-response estimates

Read:

- Figure 2 shows that the instruments for price and price interacted with aggregate elasticity are relevant.
- Table 2 reports the baseline estimate $\hat{\chi} = 2.97$ and many robustness checks.
- Most robustness estimates remain around 2.3–3.3, showing that the strategic-response estimate is stable.
- Specifications without appropriate instruments generate very different results, confirming that endogeneity matters.

Economic message: Strategic response is statistically and economically large. Ignoring simultaneity in aggregate elasticity would substantially distort the inference.

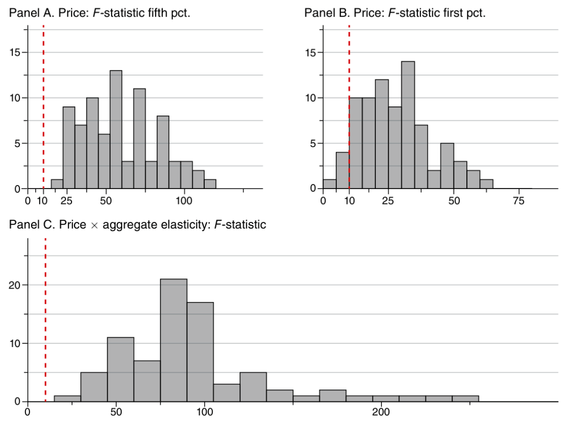

FIGURE 2. RELEVANCE CONDITION FOR THE PRICE AND ELASTICITY INSTRUMENTS

Figure 2: Instrument relevance

TABLE 2—ESTIMATES OF THE DEGREE OF STRATEGIC RESPONSE χ UNDER ALTERNATIVE SPECIFICATIONS

	Estimates for χ		Estimates for ζ	
	Estimate	SE	Estimate	SE
(1) Baseline specification	2.97	0.47	−0.48	0.14
(2) Instruments using 1yr-lagged AUM	2.71	0.49	−0.41	0.15
(3) Instruments using 2yr-lagged AUM	2.71	0.30	−0.47	0.13
(4) Instruments using 2yr investment universe	3.23	0.31	−0.54	0.25
(5) Instruments using 4yr investment universe	2.80	0.69	−0.44	0.15
(6) Additional fundamental: Book equity squared	2.84	0.33	−0.55	0.13
(7) Additional fundamental: Profitability	3.27	0.13	−0.80	0.22
(8) Additional fundamental: Investment	3.06	1.58	−0.52	0.16
(9) Additional fundamental: Dividend yield	3.01	0.35	−0.56	0.14
(10) AUM-weighted estimation	2.53	0.23	−0.33	0.25
(11) Book AUM-weighted estimation	2.65	0.41	−0.40	0.12
(12) Alternative characteristic normalization	3.01	0.30	−1.46	0.23
(13) Investor-type grouping	3.12	2.89	−0.43	0.37
(14) BE-weighted instrument for $\mathcal{E}_{agg}$	2.30	0.75	−0.31	0.44
(15) No instrument for $\mathcal{E}_{agg}$	0.04	0.22	0.47	0.06
(16) No instruments	0.63	0.29	0.47	0.16

Table 2: Estimates of strategic response

7.3 Figure 3 and Table 3: stock-level aggregate elasticities

Read:

- Figure 3 shows aggregate elasticity by market capitalization. Aggregate elasticity tends to decline with firm size.
- Table 3 summarizes average elasticity, the elasticity-size regression, and residual cross-sectional variation.
- The baseline model estimates higher average elasticity than the fixed-elasticity benchmark and more cross-sectional variation.

Economic message: Stock-level demand elasticity is not constant. It varies systematically with market capitalization and investor composition.

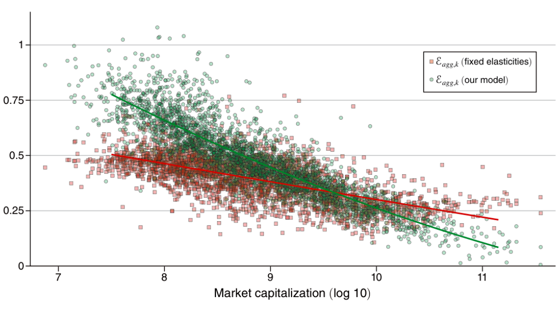

FIGURE 3. AGGREGATE ELASTICITY AT THE STOCK LEVEL: $\mathcal{E}_{agg,k}$

Figure 3: Aggregate elasticity and size

TABLE 3—PROPERTIES OF AGGREGATE ELASTICITY $\mathcal{E}_{agg}$

	Average	25th pct.	Median	75th pct.
Panel A. Statistics of average elasticity across stocks				
Elasticity $\mathcal{E}_{agg}$	0.438	0.386	0.444	0.512
Fixed elasticity	0.39	0.358	0.389	0.443
Panel B. Regression coefficient (by dates) of elasticity on size				
Elasticity $\mathcal{E}_{agg}$	−0.0777	−0.0858	−0.0728	−0.067
Fixed elasticity	−0.0286	−0.0307	−0.0273	−0.0249
Panel C. Residual cross-sectional standard deviation of elasticity				
Elasticity $\mathcal{E}_{agg}$	0.0395	0.0339	0.0376	0.0438
Fixed elasticity	0.0842	0.0739	0.0828	0.0915

Notes: Table 3 presents statistics of the aggregate elasticity $\mathcal{E}_{agg,k,t}$. We estimate the elasticities in our baseline model and in a specification with fixed elasticities ($\chi = 0$ as in Koijen and Yogo (2019)). Panel A has summary statistics of the average elasticity by date. Panel B shows summary statistics of the coefficient β_t from the regression $\mathcal{E}_{agg,k,t} = \alpha_t + \beta_t p_{k,t} + \varepsilon_{k,t}$ by date. Panel C reports summary statistics of the cross-sectional standard deviation of the residual from the regression described in panel B. The sample period is 2001–2020.

Table 3: Properties of aggregate elasticity

7.4 Figures 4 and 5: trading activity and the decline in active investors

Read:

- Figure 4 shows that trading activity is concentrated in relatively small portfolio positions.
- Figure 5 shows that the fraction of active investors falls from 81.4 percent to 59.5 percent over the sample.
- This decline is the main empirical force behind the rise of passive investing in the paper's counterfactuals.

Economic message: The decline in active investing is large, and it matters because active trading is not spread evenly across all portfolio positions.

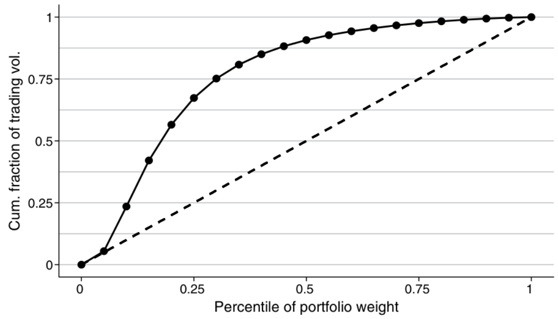

FIGURE 4. TRADING ACTIVITY ACROSS PORTFOLIO POSITIONS

Figure 4: Trading activity across portfolio positions

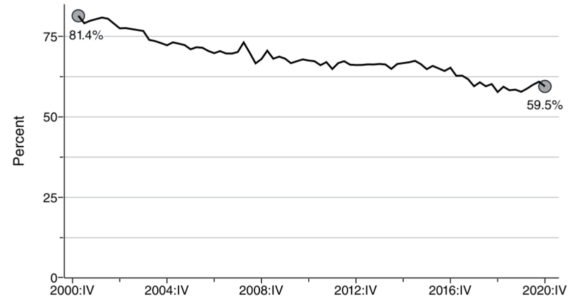

FIGURE 5. FRACTION OF ACTIVE INVESTORS

Figure 5: Fraction of active investors

7.5 Figures 6 and 7: time-series evolution and decomposition

Read:

- Figure 6 shows that the cross-sectional distribution of aggregate elasticity declines over time.
- The mean elasticity falls from roughly 0.41 to 0.27.
- Figure 7 decomposes the change into a decline in active share, a change in active investors' own elasticities, and the strategic response.
- Strategic response offsets a large part of the decline but does not fully eliminate it.

Economic message: The paper documents a secular decline in stock demand elasticity and shows that strategic response mitigates, rather than eliminates, this decline.

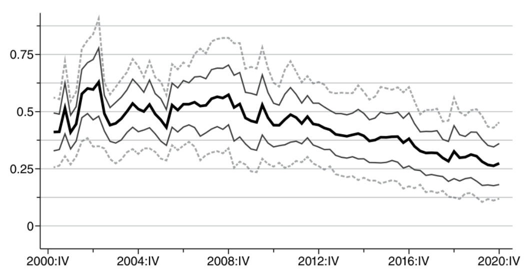

FIGURE 6. DISTRIBUTION OF AGGREGATE ELASTICITY ACROSS STOCKS

igure 6 traces out the distribution of aggregate elasticity $\mathcal{E}_{agg,k}$ over time. The bold line represents the aver-
ticity across stocks for each year. The solid lines represent the twenty-fifth and seventy-fifth percentiles, and
ied lines the tenth and ninetieth percentiles.

Figure 6: Distribution of aggregate elasticity

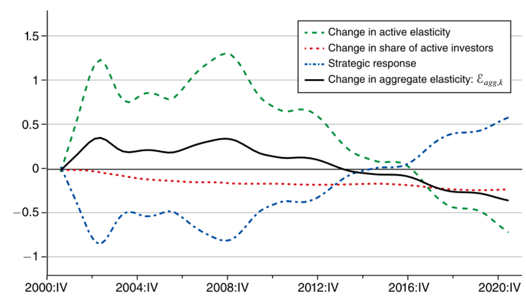

FIGURE 7. DECOMPOSITION OF THE CHANGE IN AGGREGATE ELASTICITY

igure 7 shows the decomposition derived in equation (29) over time. We compute each term of the decom-

Figure 7: Decomposition of elasticity changes

7.6 Figure 8 and Table 4: counterfactual competition and active share pass-through

Read:

- Figure 8 compares the baseline elasticity path to two counterfactual regimes: full strategic response and no strategic response.
- With full response, aggregate elasticity barely falls. With no response, elasticity falls much more sharply.
- Table 4 shows that stock-level changes in active share predict changes in aggregate elasticity with a coefficient near the theoretical pass-through of one-third.

Economic message: Strategic response is the main reason passive investing has not reduced aggregate elasticity even more.

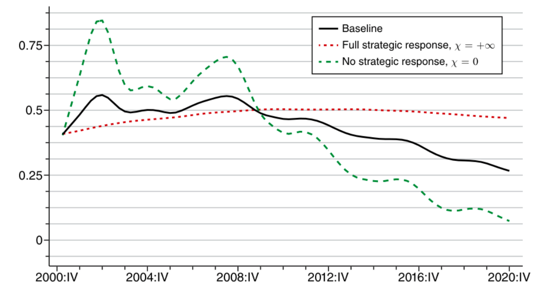

FIGURE 8. THE EVOLUTION OF AGGREGATE ELASTICITY UNDER ALTERNATIVE COMPETITION REGIMES

Figure 8 shows the evolution of aggregate elasticity $\mathcal{E}_{agg,k}$ under alternative strategic regimes. The bold

Figure 8: Alternative competition regimes

TABLE 4—CHANGES IN AGGREGATE STOCK-LEVEL ELASTICITY $\mathcal{E}_{agg,k}$ ON THE ACTIVE SHARE

	log change in elasticity				
	(1)	(2)	(3)	(4)	(5)
Change in active share	0.259	0.385	0.368	0.346	0.336
	(0.059)	(0.022)	(0.022)	(0.021)	(0.055)
Date fixed effects		Yes	Yes	Yes	Yes
Stock fixed effects			Yes		
Controls				Yes	Yes
Estimator	OLS	OLS	OLS	OLS	IV
Observations	50,292	50,292	49,661	50,292	10,619
R^2	0.028	0.612	0.637	0.695	0.696
First-stage F-statistic					9.754
First-stage p-value					0.000

Notes: Table 4 reports a panel regression of annual log change in stock-level elasticity $\mathcal{E}_{agg,k}$ on the annual log change in the active share $|Active_k|$. Column 2 adds date fixed effects. Column 3 adds stock fixed effects. Column 4 uses date

Table 4: Active share and elasticity changes

7.7 Table 5: elasticity and asset-price behavior

Read:

- Table 5 relates stock-level aggregate elasticity to volatility, price informativeness, and illiquidity.
- More elastic demand is generally associated with lower volatility.
- The evidence for price informativeness is weaker, while illiquidity relations vary across OLS and IV specifications.
- The table shows that aggregate elasticity has economically meaningful links to asset-market outcomes.

Economic message: The estimated demand elasticities are not just accounting objects. They help explain cross-sectional differences in price behavior.

TABLE 5—STOCK-LEVEL ELASTICITIES $\mathcal{E}_{agg,k}$ AND THE BEHAVIOR OF ASSET PRICES

	Total volatility		Idiosyncratic volatility		Price informativeness		Illiquidity	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Elasticity	−0.059	−0.383	−0.034	−0.288	0.019	0.258	0.750	−0.396
	(0.049)	(0.123)	(0.041)	(0.100)	(0.186)	(0.474)	(0.067)	(0.131)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Estimator	OLS	IV	OLS	IV	OLS	IV	OLS	IV
Observations	222,359	222,359	222,261	222,261	67,376	67,376	219,780	219,780
R^2	0.218	0.208	0.248	0.240	0.020	0.018	0.714	0.619

Notes: Table 5 reports panel regressions of measures of volatility, price informativeness, and illiquidity on stock-level elasticity $\mathcal{E}_{agg,k}$. All variables are de-meaned and standardized for each date. Odd columns show results from OLS regressions. Even columns show results from instrumental variables regressions that use our instrument for stock elasticity defined in equation (24). For columns 1 and 2, we compute the total daily volatility of stocks from daily return data (Center for Research in Security Prices 1999–2020). For columns 3 and 4, we compute daily idiosyncratic volatility with respect to the Fama and French (1993) three-factor model based on daily CRSP data (Center for Research in Security Prices 1999–2020) and daily factor return data (French 1999–2020) within a quarter. Columns 5 and 6 take the measure of price informativeness provided by Dávila and Parlatore (2018a,b). Columns 7 and 8 use the Amihud (2002) measure for illiquidity as the dependent variable, calculated again based on daily CRSP data (Center for Research in Security Prices 1999–2020) within a quarter. All specifications are weighted by lagged market equity. We follow our main specification for the estimation of elasticity and control nonlinearly for book equity. The sample period starts in 2001 for all columns and ends in 2020 for spec-

Table 5: Stock-level elasticities and asset prices

8 Short summary

- The paper estimates how much investors change their trading aggressiveness when other investors become less aggressive.
- The key parameter is χ, the degree of strategic response.
- The baseline estimate is $\chi = 2.97$, implying substantial but incomplete competition.

- The growth of passive investing lowers aggregate stock demand elasticity because active investors only partially offset the decline in active trading.
- The authors estimate that passive investing made individual-stock demand curves 11 percent more inelastic over 2001–2020.
- Cross-sectional variation in investor mixes solves the reflection problem behind strategic response estimation.
- Aggregate elasticity is lower for large stocks and has meaningful relations with volatility, informativeness, and illiquidity.

9 Conclusion

9.1 Core conclusion

Haddad, Huebner, and Loualiche show that the US stock market is competitive, but not perfectly so. Active investors do respond when other investors become less aggressive, yet the response is incomplete. As a result, changes in investor composition, especially the rise of passive investing, affect stock-level demand elasticities and price behavior.

9.2 Economic intuition

When passive investors hold a larger share of a stock, less price-elastic demand is available. Active investors notice that fewer traders are exploiting mispricing and become more aggressive. However, information costs, mandates, incentives, cognitive frictions, and limits to arbitrage prevent them from fully replacing the lost elasticity. The outcome is a partially competitive stock market.

9.3 Takeaway

The paper's main message is that market organization matters. The identity and behavior of investors affect aggregate demand curves, which in turn affect prices. Passive investing is not neutral, but its impact is mitigated by strategic responses from active investors.